



Community detection in social recommender systems: a survey

Fabio Gasparetti¹ · Giuseppe Sansonetti¹ · Alessandro Micarelli¹

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Abstract

Information extracted from social network services promise to improve the accuracy of recommender systems in various domains. Against this background, community detection techniques help us understand more of users' collective behavior by clustering similar users w.r.t. their interests, preferences and activities. The purpose of this paper is to bring the novice or practitioner quickly up to date with the main outcomes and research directions in the field of social recommendation based on community detection. The research synthesis consists of a narrative review which identifies what has been written on the topic of community-based recommender system. The comprehensive search of relevant literature aims at synthesizing prior study findings by identifying approaches that follow similar paradigms and techniques. The paper is of value to those involved with recommender systems and social media.

Keywords Community detection · Recommender systems · Social network services

1 Introduction

Popular online social networks, such as Facebook or Twitter, consist of nodes representing people or other entities embedded in a social context, and edges referring to interactions, collaborations, or some other form of linkage between users and entities. The power and appeal of social networking are such that the number of active users worldwide is continuously increasing [116].

Meanwhile Recommender systems (RS) have been gathering increasingly crucial role and influence on our daily life. They are implemented in many web applications for guiding user choices, such as online retailers for recommending items to buy or news articles to read. RS are mostly based on representations of user preferences and interests. Social information, such as contacts and interactions between peers provide a large breadth of signals useful in identifying these aspects of the users. Indeed, research activity on *Social recommender systems* (SRS) focuses on tapping social information in the ranking process [50], in order to provide more accurate and unexpected recommendations,

potentially more efficiently, with respect to more traditional approaches.

For instance, Collaborative filtering (CF) based recommenders are often based on user-user similarity measures which can be improved by analyzing the social relationships in the online networks (see Section 4.4). The common rationale is that the likelihood of accepting people's advice is affected by tie strength, trustworthiness, and homophily towards them [4]. Hence, trusted recommendations from people with similar tastes are considered more than those from strangers and vendors.

Indeed, representations of user preferences and interests may be enhanced by analyzing the characteristics of both the user's *egocentric network* (i.e., the network or ties around an individual) and the *sociocentric network* (i.e., different patterns of interaction within defined groups and communities).

Community-based recommendation consists of peculiar techniques that explore the network structures and social cues to perform recommendations. The assumption is that the problem of effective recommendation can be addressed not exclusively by considering the egocentric network or single users with similar preferences. Instead, groups of users that have more similar social characteristics and typically strong ties than might normally be encountered in the rest of the network (i.e., a *community*) may provide precious and additional evidence for the recommendation

✉ Fabio Gasparetti
gaspare@dia.uniroma3.it

¹ Roma Tre University, Rome, Italy

process. Hence, they are good at obtaining and sharing resources between peers that have common characteristics with the target user, even if the user does not have any direct relationship with the peers.

The large complexity of social networks and the heterogeneity of the shared elements provide us abundant information to learn from but may also hinder the accuracy of SRS. The partitioning of a social network into densely tied subgroups is important in understanding the likely behavior of each community and the network as whole. The available information may be categorized and conveyed in such a way that each user is less likely to miss important facts because he is struggling to process all the information coming in.

Besides improving the accuracy, the classification of the users based on their role with respect to the communities they belong to, has the chance to prevent the *cold start problem* [115], which frequently occurs in dynamic and heterogeneous environments. It is caused by items that cannot be recommended until sufficient information to represent them has been acquired, e.g., ratings, opinions, number of online purchases.

RS that take advantage of social analyses of virtually unbounded networks face serious scalability problems. Large amounts of operational data originate every moment and mapping all the potential relationships between peers is inherently impossible. Community detection splits the network down to several small-scale groups where more traditional and resource-intensive recommendation techniques can be implemented. The computational complexity of traditional recommendation algorithms directly depends upon the number of existing users and entities in the network. Growth in number of these factors results in either slow down or requires more resources. By limiting the computation on specific communities of users of interest, the resource can significantly be reduced [72].

In this paper, a systematic narrative review synthesizes the research evidence from individual studies proving how community detection techniques have the chance to provide benefits in the recommendation process. Specifically, we report research studies on social recommender systems that take advantage of community detection to face one of more of these challenges:

- To tap additional information from the sociocentric network of the user.
- To shrink the amount of less relevant information with respect to the recommendation task.
- To address the cold start problem.
- To limit the computational complexity of the recommendation algorithms and guarantee their scalability.

The approaches discussed in this survey are based on a variety of technologies, spanning from traditional collaborative

filtering and content-based recommender systems to machine learning and social network analysis. Wherever possible, each technology will be briefly introduced, referring to technical literature for details. Hence, this survey is intended to practitioners but accessible to readers with basic notions of network science.

This survey is focused on recommender systems that make use of community detection algorithms in order to provide better outcomes and guarantee the scalability on large and complex domains. Section 3 cites the principal community detection approaches, with references to those that explicitly address the challenge to analyze very large graphs, such as in the social network services. Azaouzi et al. [6] provide a comprehensive and recent survey of community detection algorithms in social graphs.

For convenience, Table 1 reports recent surveys on methods, approaches, theories and analyses of datasets to determine features that are considered relevant to the domain under discussion. None of the them explicitly include overviews on recommender systems that take advantage of community detection or graph clustering approaches, or discuss their underlying rationale, benefits and challenges. A very few have some references in common with respect to specific topics, such as serendipity, diversity and scalability. However, several of these surveys (e.g., [28, 64, 67, 93, 113]) are extremely relevant for investigating domains and techniques that are often used in combination with recommender systems and social network services. The most popular of these techniques are listed for convenience in Table 2.

The first section of this manuscript introduces a formulation of the recommendation task on social networks. Community detection in large graphs is introduced in Section 3. Section 4 hosts the discussion on several aspects, techniques and methodologies related to community-based social recommendation. In particular, data collection and content extraction (Section 4.1), the inference of explicit and implicit social ties (Section 4.2), dimensionality reduction techniques (Section 4.3) and, finally, a review of the literature in Section 4.5. Concluding remarks and major research challenges of Section 5 close the work.

2 Problem formulation

The formulation of the recommendation problem can be cast in the *link prediction* task [80]. A graph $G = (V, E)$ represents relevant elements from a given social network, e.g., individuals and objects of interest. So, in general it consists of several types of vertices $V = V_1 \cup V_2 \cup \dots \cup V_{N_V}$ and edges $E = E_1 \cup E_2 \cup \dots \cup E_{N_E}$, such that the set V_i includes the vertices of the i -type and, similarly, E_j consists of the edges of the j -type. For the sake of convenience, the

Table 1 Recent surveys on methods, approaches, theories and features relevant to the discussion

References	Year	Topic	Domains of Interest	Intro to traditional RS techniques	Focus on community-based RSs	Review of literature	Overlap	Goal
Aggarwal [2]	2016	Trust in SRS	Any	Partial	No	Yes	Trust	Review of literature
Dou et al. [34]	2016	CF & SRS	Any	Yes	No	Yes	No	Short survey on CF for SRS
Kotkov et al. [70]	2016	Serendipity	Any	No	No	Yes	Serendipity	Review of literature
Khan et al. [67]	2017	Cross-domain RSs	Any	No	No	Yes	No	Review of literature
Kunaver and Požžrl [71]	2017	Diversity	Any	Partial	No	Yes	Diversity	Review of literature
Karim et al. [64]	2018	News RSs	News	No	No	Yes	Scalability	Review of literature and challenges
Mu [93]	2018	DL & RSs	Any	Partial	No	Yes	No	Review of literature
Azaouzi et al. [6]	2019	CD in SNS	Any	No	No	Yes	No	Review of literature
Bai et al. [8]	2019	RSs	Scholar	Yes	No	Yes	No	Review of literature
Jannach et al. [62]	2019	Session-based RSs	e-com	No	No	No	No	Feature-based analysis & new algorithms
Sahoo et al. [110]	2019	Health RSs	Healthcare	Yes	No	Yes	No	Review of literature & new algorithms
Dara et al. [28]	2020	Group RSs	Any	No	No	Yes	No	Review of literature
Shokeen and Rana [113]	2020	SRS	Any	Yes	No	Yes	No	Investigates SNS features for RSs

(CD) Community detection algorithms, (CF) Collaborative filtering, (DL) Deep Learning architectures, (e-com) e-commerce, (RS) Recommender Systems, (SNS) Social Network Services, (SRS) Social Recommender Systems

Table 2 Legend of the principal algorithms and techniques referenced by the surveyed community-based recommender systems, with references to additional literature

	Acronyms	Sections	Refs.
Techniques used for Community detection			
Community detection based on Multi-agent system	ACD	Section 4.5.2 (Temporal aspects)	[1]
Community detection based on dense bipartite graph	DBG	Section 4.5.1	[107]
Communities corresponds to social network's groups	EG	Section 4.5.2 (Multiple affiliation problem)	[125]
Modularity-based community detection	LCD	Section 3	[6]
Latent Dirichlet Allocation	LDA	Section 4.5.2 (Latent models)	[10]
Local Fitness Maximization	LFM	Section 4.5.2 (Temporal aspects)	[73]
pLSA-based model	pLSA	Section 4.5.2 (Latent models)	[55]
Singular value decomposition	SVD	Section 4.5.2 (Latent models)	[35]
Common unsupervised clustering algorithm	UC	Section 4.5.1	–
Techniques for the Ranking of the recommended items			
Association rule-based recommendation	AR	Section 4.5.2 (Temporal aspects)	[3]
Content-based ranking	CB	Section 4.5.2 (Diversity and serendipity)	[45]
User-user similarity based on Kullback–Leibler divergence	KL	Section 4.5.2 (Latent models)	[26]
User-user similarity based on Jensen–Shannon divergence	JS	Section 4.5.2 (Latent models)	[91]
Linear combination	LC	Section 4.5.1 and 4.5.2	–
Matrix factorization	MF	Section 4.3 and 4.5.2 (Latent models)	[69]
Multi-armed bandit model	MAB	Section 4.5.2 (Multiple affiliation problem)	[65]
Sentiment analysis-based recommendation	SA	Section 4.5.1 (Estimating the tie strength)	[54, 79, 103]
Ranking based on user-item tie strength	TS	Section 4.5.2 (Multiple affiliation problem)	[125]

set $V_U \subset V$ indicates the users, and $V_I \subset V$ the objects to be recommended.

By considering the most popular social network services, the E set typically includes family members and friendships, or following/follower relationships. Since these platforms tend to let users make reference or upload a wide variety of external resources, such as videos, web pages and comments; in order to keep the formulation simple, we name these objects *items*. The inclusion of this heterogeneous set of items makes the social graph multipartite according to the available categories to which each vertex can be partitioned into.

The user-item interaction represents a specific action taken by the user on the item, e.g., sharing, liking or leaving a comment. Common e-commerce online services and crowd-sourced review websites (e.g., Yelp, Tripadvisor) provide the users with way to leaves a trail of the user-item experience, for instance, by submitting ratings or reviews after the purchase or a point-of-interest visit; and in some circumstances relationships between users that might happen when one user is interested in following the activities of others.

Figure 1 shows a snippet of a huge graph built by considering the Twitter social network. Two categories of vertices are represented, namely, Twitter users and posted microposts (or *tweets*). Different types of edges connect

the graph's nodes including traditional social ties between users, and the user interactions with the published tweets.

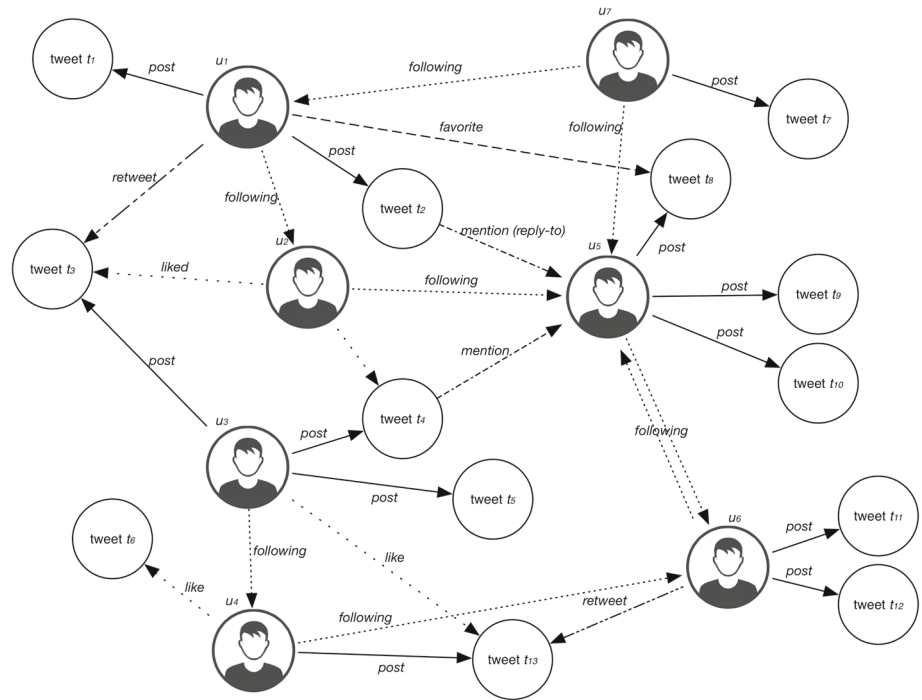
By clustering the vertices in G in communities (or *partitions*) of nodes with similar characteristics we obtain a partition where each vertex $v_k \in V$ can be associated with one or more sets in $C = \{C_1, C_2, \dots, C_{N_C}\}$, each representing a community.

Popular online social networks provide specific interface' functionalities to individuals for explicitly building virtual communities and sharing their common interests and express their opinions, such as *Facebook groups* and *Twitter lists*. However, partitions of users with similar characteristics can be also inferred by community detection algorithms by mining large datasets of usage data. Figure 2a shows a potential partition obtained from the initial snippet of the network by considering solely the follow relationships between peers.

The link prediction task consists of predicting whether, given a particular category of edges E_j and two vertices $(v_k, v_l) \notin E_j$, an edge of j -type will connect them in the future. Four macro-categories of links are usually considered:

People to people (P2P) It supports the discovery and establishment of social ties among existing users or newcomers. In the snippet on Fig. 1, the user u_3 may be interested

Fig. 1 A graphical representation of a collection of objects and types of relationships in the case of Twitter social network



in following the user u_6 since u_3 mentioned u_6 in a post and also liked a post retweeted by u_6 . This specific task is often named *people-to-people recommendation*.

People to community (P2C) It represents the level of membership of one user to a specific community. Whereas it is fundamental for any kind of community-based recommender, it is strategic for the formation of teams and communities of experts and for increasing their cohesiveness.

Content to people (I2P) The category refers to the traditional recommendation, where selected classes of items (e.g. house furniture, point of interests) have to be sifted through to build lists of suggestions for a target user. In Fig. 1 an instance of the I2P link prediction aims at inferring the *like* category of edges. For instance, u_3 may be recommended with the post t_6 , which has been liked by the followed user u_4 .

Content to community (I2C) The last category is a specific case of the previous recommendation, where the top-ranked items reflects the interests and preferences of all group members.

The categories P2C and I2C often refer to *group recommendation*. They show peculiar complexity since they address circumstances in which individuals show conflicting tastes and their preferences may change due to the presence of others in the group. Hence, specific aggregation strategies for the generation of a common consensus point are required. For the sake of conciseness, this survey is focused on P2P and I2P recommendations,

and we refer to other reports for in-depth analyses of group recommendation and its challenges [14, 28].

Under these settings, our goal is to implement the following utility function r :

$$r(G, C, v_k, v_l, j) \rightarrow [0, 1] \quad (1)$$

that aims at estimating the likelihood of the existence of a link of j -type between the v_k and v_l nodes in V , given the graph G and the identified communities C . If the V_i is the subset of V that includes the items of interest for the recommendation, and v_k is the vertex corresponding to the target user, the chosen item that is to be suggested is obtained as follows:

$$\arg \max_{v_l \in V_i} r(G, C, v_k, v_l, j) \quad (2)$$

The predicted value of interest allows us also to build up a ranked list of items that maximize the utility function so that a given performance measure is maximized. Most of the times this performance measure corresponds to the estimate of satisfaction of the users of the recommender system, but it may also refers to the overall level of interaction. Indeed, social network services aim at increasing the density of interactions of their members, so that the perception of utility of their use is certain.

In the I2P scenario, we have $v_k \in V_U$, $v_l \in V_I$ and the edge usually corresponds to actions such as *buying* or *watching* the specific item. While one vertex is usually represented by static characteristics (e.g., the genre of a book, the segment of a car), in the P2P scenario both sides are

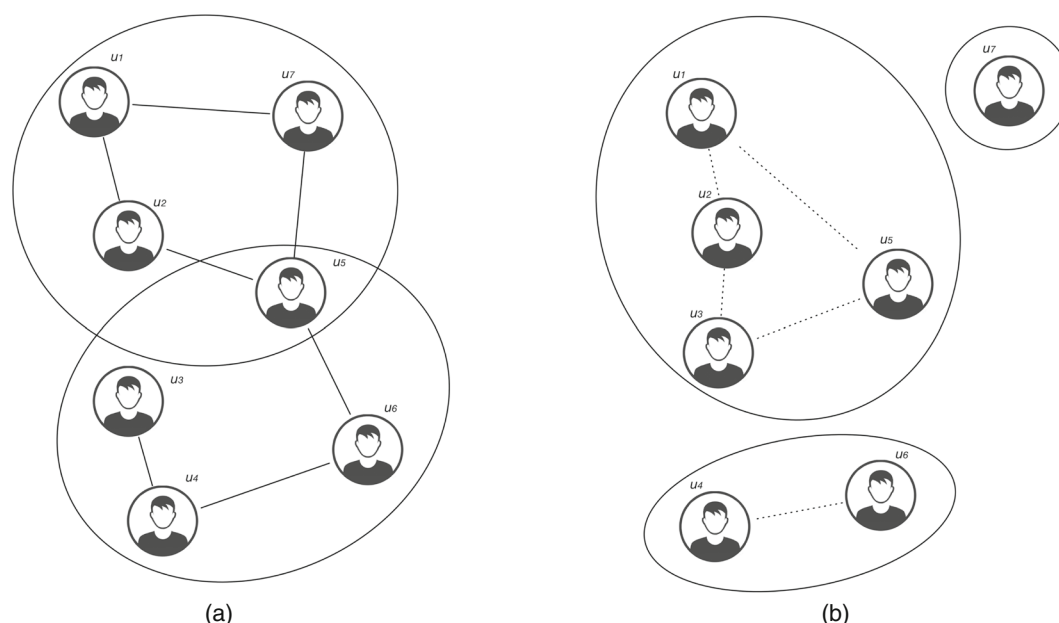


Fig. 2 Two partitions of a social network, one obtained by considering explicit ties (a), and one analyzing implicit ties (b). Hence, the identified communities heavily depend on the considered correlations that bind two users, so the same user might belong to different

users who actively participate in social interactions. A typical instance of a P2P recommendation consists in *online dating* websites, that provide the means for people to “meet” others virtually, and a relationship is considered relevant when both parties are interested in each other [105]. This kind of *reciprocal* recommendation is considered more challenging, since both sides of user preferences and taste shall be considered, and they may shift at any given moment.

Before reporting on the principal stages of a social recommender systems in Section 4, a brief mention of community detection algorithms for large web graphs is included in the next section.

3 Community detection in large networks

Community detection is also called *graph* or *network clustering* [44]. In our context its goal is to partition large networks of users into subgraphs.

More formally, the *intra-cluster density* of the subgraph G' can be obtained by calculating the ratio between the number of internal edges of G' and the number of all possible internal edges. The *inter-cluster density* is the ratio between the number of edges running from the vertices of G' to the rest of the graph G and the maximum number of potential inter-cluster edges. The optimal partition maximizes the differences between the intra-cluster and inter-cluster density over all the communities. In other words, nodes of the same group are tightly interconnected,

whereas looser connections bind nodes from distinct groups. Similarly, *modularity* measures indicate quality indices in partitioning a network into communities. High modularity corresponds to configurations with high intra-cluster and low inter-cluster density.

Consequently for several types of explicit and implicit connections that bind users and items in large communities, it is not easy to develop a unified framework of community detection that takes into account all of them. So, multipartite graphs are usually reduced to synthetic unipartite projections of specific vertex and edge classes. A typical scenario consists of vertexes and edges that represent users and friendship ties, respectively. Each vertex may be also associated with categorical or numerical *attributes* that describe specific characteristics, such as the interests or the preferences of each individual, and with partitions that group individuals with similar attributes. Implicit relations between vertexes may be also inferred, as discussed in Section 4.2.

Girvan and Newman [47, 99] propose one of the first algorithms for community detection for small networks. It is a *divisive* (or top-down) method in which edges are identified and progressively removed from a network by computing the *edge betweenness* score. It corresponds with the number of shortest paths between vertex pairs that run along the edge in consideration, summed over all vertex pairs. The algorithm iteratively calculates the betweenness of all edges in the network and removes the one with highest score, and repeats the process until no edges remain,

following a *greedy* paradigm. Its underlying principle calls for removing the edges that connect different communities.

Most of the community detection algorithms can be classified into two categories: based on heuristics or objective function optimization algorithms. The detection task of searching for the optimal modularity value has high complexity (NP-hard). For instance, the Girvan and Newman algorithm has $O(m^2n)$ time complexity on an arbitrary graph with m edges and n vertices, and $O(n^3)$ on a sparse graph. Indeed, in an online social network scenario, the space of potential partitions grows faster than any power of the system size. A heuristic search strategy is mandatory to restrict the search space while preserving the optimization goal for large and complex networks. Thus most of the approaches of our interest aim at combining specific heuristics in the optimization process.

A popular approach for large networks has been proposed by Newman [98]. It is a modularity-optimizing community detection algorithm, which is still based on a standard greedy paradigm but is capable of pruning the search space of the less likely configurations. Based on the Newman's algorithm, Clauset et al. [25] propose a fast greedy optimization. It integrates specific heuristics and sophisticated data structures so that the running time of the algorithm is essentially linear with the number of vertices.

The algorithm proposed in Duch and Arenas [36] begins with a random bipartition and use each node's contribution to the modularity as a fitness test. At each iteration it moves the node with the lowest fitness to the opposite partition. It makes use of a heuristic search based on the Extremal Optimization (EO) algorithm proposed in [13]. It successively replaces extremely undesirable variables, which represent poorest degree of adaptation to the optimal value, with new values drawn randomly from a flat distribution. This heuristic has proved to be more effective w.r.t. traditional simulated annealing and genetic algorithms.

The popular Louvain algorithm [11] finds high modularity partitions and that unfolds a complete hierarchical community structure for the network, thereby giving access to different resolutions of the communities. Computer simulations on large ad-hoc networks suggest that its complexity is linear on typical and sparse data. Additional function optimization algorithms for community detection on large networks make use of approximation approaches, e.g., see [51, 84].

The assumption of most of the community detection algorithms is that the whole network is available for the analysis. However, this rarely happens in the scenario of social network services. Several online platforms or third-party services (e.g., Datasift [31], Gnip [59] now part of Twitter APIs, and Socialgist [114]) provide programmatic access to both the published data and user information in the

form of REST APIs (Application Programming Interface). As most of the online services, microblogging platforms and external services follow cost models that guarantee free access to a limited amount of the published data. Moreover, privacy constraints do not allow the discovery of the entire structure and its elements by external services, as is the case of Facebook. For these reasons, community detection techniques can also operate by sampling the large graph-based representation of the available network [24, 43, 75, 83, 95, 108], though these methods assume knowledge of the structure of the entire network for the sampling process.

Since the content published on social networks is continuously updated by their users, there are currently a lot of activities on the development of methods to analyze temporal networks for detecting dynamic communities. Of course traditional algorithms used for static graphs can be used for dynamic networks by considering a succession of snapshots $G_{t_1}, G_{t_2}, \dots$ one at a time. More advanced approaches analyze the effects that small variations in the network structure has on the graph and, therefore, adjusting the partition of the already-computed configurations obtained in the previous snapshot [7, 21, 23]. A key advantage is that evidence acquired in previous stages can be systematically exploited to provide computation/optimality trade-off.

In terms of specific applications of community detection algorithms on large graphs obtained by social networks, Fan and Yeung collect data from the Facebook platform in the form of friendship relationships, like actions and information such as two users live in the same city [38]. The authors adapt the traditional Newman's modularity optimization so that additional networks that encode different information with respect to friendships (e.g., likes and common hometowns) are exploited to improve the accuracy of the obtained communities. An original agent-based computation approach for distributed networks iteratively determines communities by series of discrete evolutionary cycles [16]. The approach is particularly suitable to keep a balance between high modularity and short running time. Snapshots of Facebook graph have been considered in the experiments to prove the efficacy of the algorithm. For a complete overview, Azaouzi et al. [6] report recent advancements in community detection in large-scale social networks.

Users in the same community are expected to more likely share similar temporal behavior than others. Shaping Twitter communities in a way to group together users who show similar temporal behavior towards the topics of interest was the goal pursued in [40]. The authors devise a space of dense low-dimensional vectors, namely *embeddings*, where users who have similar inclination towards similar topics in similar time intervals will be represented close to each other. Further approaches that take into account

temporal dimension make use of a multivariate timeseries representation of users in topic and time spaces [39], or unified probabilistic generative models of both topics and users [58].

4 Recommender systems based on communities

Leveraging communities identified by analyzing large social networks in the recommendation process is complex and requires multiple processing stages, as summarized in Fig. 3.

After having collected enough information for representing social relationships, interests, preferences and activities of the users, the tie recognition stage exploits this data to formalize the user-user and user-item connections. The system analyzes these ties to discover communities of users having similar tastes. Dimensionality reduction is often considered for reducing data sparsity. Details of each stage are discussed in the following sections.

4.1 Data collection, content extraction and enrichment

The first step interacts with the online service in order to collect relevant information. Data is collected by aggregating all the observed user-user and user-item interactions in the network over a period of time. The output is a single graph G .

In order to provide accurate recommendations, RS usually collects additional information about users by performing content extraction and enrichment. Typically, this information takes form of attributes representing demographic features (e.g., income level, age, geographic location, occupation) and interests for which a recommendation is sought (e.g., hobbies, favorite books and musical tastes).

In social networks, users are often characterized by their past history of activities in terms of published content. Let us take a scenario, where the user shares a news article on the issue of plastics in the ocean. If we are able to derive concepts such environmental activism, protecting biodiversity and sustainability; we can better characterize the user interests and goals. In a similar way, customers of e-commerce services are linked to the products they have rated or bought in the past. Customer classification is based on the characteristics of products they purchased.

Common features are topics, tags and concepts extracted from shared text documents and music genres assigned to songs listened to. This is a typical instance of *user modeling*, where explicit representations of some characteristics of the users are acquired during their normal activities on the online platform [46]. Typically, the output of the content extraction is an attributed graph where each extracted feature is represented by a node attribute. Equivalent representations are based on user-item matrices that may represent user activities such as purchases or ratings.

For instance, Tchuente et al. [117] explicitly represent user profiles with traditional vectors in a high dimensional space [46], where each dimension (or attribute) denotes the membership level to one or more communities. An external taxonomy provides a hierarchical organization of the topics associated with each community, so that profiles have different granularity levels, e.g., Sports $\rightarrow$ Air Sports $\rightarrow$ Parachuting $\rightarrow$ Freeflying.

4.2 Social tie inference

Since, ties between peers are a fundamental source for evaluating similarities between users and identifying significant communities, it is appropriate defining two sub-categories of ties.

Explicit ties are usually gathered during data collection and can be interpreted as trust relationships between users, since they are explicitly stated by them on the social platform. For instance, the Twitter's *follow* relationship is defined by the user who wants to keep abreast of the posts of the followed peer. Explicit ties can also take the form of user-item relationships. For instance, when the user publishes, rates, comments or purchase an item on the network.

Since, users are often similar to one another even if no explicit relationship exists between them, further types of similarities take into account the interactive behavior and the model of the users. In particular, two users might be judged similar if they have similar interests or attributes (*profile-based similarity*), or if they show a similar behavior, such as rate or like content in a similar way (*rating-based similarity*). This sort of similarities establishes *implicit* or *weak ties* between users because there is not any direct interaction. The tie is mediated by a representation of specific features or profiles of the users, which do not exactly correspond to resources shared in the network. Figure 2b shows a distinct partition obtained from the same initial snippet by solely considering implicit ties.



Fig. 3 Steps in the recommendation based on communities

Implicit ties have the fundamental benefit of establishing alternative paths between users and items that can be exploited for the recommendation and, therefore, ensuring that the analysis of the network satisfies the *small world* characteristic [68]. In this circumstance, the average path length grows logarithmically as a function of the number of nodes in the network. In other words, most of the nodes can be reached from every other node by a small number of steps. These *implied* networks are characterized by having a small diameter and high clustering, backing the effectiveness of the community detection techniques. In some circumstances, graphs of implicit ties are also useful for addressing the cold start problem [115], and are also considered as an important source for diversified information in virtual communities (e.g., see [131]).

The principal drawbacks of collecting implicit ties from social networks are the following:

- Most of the times negative examples are missing. If the users do not like horror TV shows, they probably do not leave any trace on the network. Similarly, if a link between one user and the horror show is missing, it is not clear if the user does not like the show genre, or the positive feedback has not been left by the user yet. In addition, link prediction based on supervised learning techniques that require a similar number of positive and negative training instances call for adequate techniques for solving the class imbalance problem [101, 106].
- An implicit feedback does not have associated a numerical value of preference, such as ratings or votes. However, it is possible to devise a strength measure that depends on the kind of user-user or user-item interaction, which is combination of multiple factors, such as, the frequency and time-span of the interactions. This measure is often named *tie strength* [49, 76], which indicates a confidence that we have in a certain observation. For instance, by counting the times the user listened to a song the system might deduce how much she likes its author.
- Implicit feedbacks are inherently noisy since their inference from the user behavior is not always accurate.

4.3 Dimensionality reduction

Collaborative filtering [48], one of the most popular recommendation techniques, uses the feedback from each individual user. It relies on *likes* and *dislikes* expressed by the user on items, which are exploited for computing a similarity index to build a ranked list of suggested resources. These ratings form a large matrix R of values in which the users are the rows, the items the columns, and the values in the cells represent the known preferences expressed by

users on items. CF paradigm underlies most of the largest e-commerce platforms available online [12].

The R matrix contains many missing values since users usually interact with a small portion of items in the particular application domain. For instance, in the movie domain, a matrix built on a real-world dataset of user-item ratings (namely, MovieLens dataset [53]) has no information for 90% of its elements. With such a *data sparsity*, traditional CF-based approaches do not always have means to estimate similarity measures between users or items.

Latent Semantic Analysis (LSA) [32] and Matrix Factorization (MF) [69] are models that map the initial high-dimensional matrix R into a f -dimensional (low-dimensional) space. The underlying idea is to map users and items into a space of latent dimensions, and use them to perform link prediction. So, different items with very similar characteristics may be assigned to the same latent representation. *Model-based* CF uses these latent features of users and items to learn a predictive model and infer the unknown preferences.

Besides potentially reducing the data sparsity issue, this form of dimensionality reduction provides two major benefits in the recommendation process, that is:

- Scalability: Mapping the initial dataset to a lower-dimensional space is a compute-intensive process, and it is usually more difficult to add data to model-based systems, making them inflexible. But usually the obtained models are much more compact than the initial dataset, so that they can be used efficiently even if the number of users and items are very large. For this reason, model-based CF is able to scale well to real-world applications [74].
- Fast predictions: Besides the computational time required to obtain the model, the query execution time is much smaller because of the compact dimensions.

MF techniques are often considered for learning the latent characteristics of users and resources in the network. Approximations of the r function (see (1)) can be defined by means of inner products in that latent space. More formally, the predicted ratings $\hat{R} \in \mathbb{R}^{|V_U| \times |V_I|}$ are modeled as follows:

$$\hat{R} = r_M + QP^T \quad (3)$$

where $P \in \mathbb{R}^{|V_U| \times f}$ and $Q \in \mathbb{R}^{|V_I| \times f}$, f is the number of dimension of the latent space chosen so that $f \ll |V_I|, |V_U|$, and $r_M \in \mathbb{R}$ is a global offset. P and Q essentially represents the latent representations of user profiles and items, respectively.

The observed rating values are considered for deriving the two matrices P and Q , characterized by vectors p_* and q_* that minimize the following regularized squared error:

$$\min_{q^*, p^*} \sum_{v_k \in V_U, v_l \in V_I} (r_{k,l} - \hat{r}_{k,l})^2 + \lambda (\|p_k\|^2 + \|q_l\|^2) \quad (4)$$

where the latter term and the $\lambda > 0$ parameter regularize the learned matrices P and Q as to prevent over-fitting.

The value $r_{k,l}$ corresponds to the actual known preference, whereas $\hat{r}_{k,l}$ is the value predicted by the model in (3) for the k -user to l -item obtained by the expression $q_l p_k^T$. An iterative approach based on the *alternating least squares* technique with regularization may be adopted for ensuring the convergence of the (4) [57], that is, when either the matrices P and Q are no longer changing or the change is not significant.

Whereas MF has been successfully applied to user-rating matrices, Section 4.5.2 makes reference to MF approaches that use also additional information (e.g., social ties and attributes) collected from social platforms.

4.4 Social recommendation

People tend to relate to other people with similar attributes. Consequently, people who have close ties tend to influence each other and have similar traits [122]. These phenomena are the foundations of the SRS, which take user interests and trusts manifested on the social platforms as additional inputs [50]. As most of the current recommendations are primarily based on the entire social graph, SRS exploit

relationships following the hypothesis that users' familiarity network strongly affects their preferences.

Neighborhood-based CF is a *memory-based* approach for predicting the preference that the target user u may assign to a given item l by storing all the preferences in memory. Like many recommender systems, it follows a two-phase paradigm. Initially, a group of users $\mathbb{H}_u$ (or *neighborhood* set) who have sufficient data for appropriate application are collected, and, subsequently, the individuals' preferences and judgments are aggregated to output the recommendation result [77, 102]. Figure 4a shows a neighborhood for the u_4 user.

In the traditional CF, the $\mathbb{H}_u$ set is obtained by identifying the users most similar to u who have rated the target item l . More formally, we introduce the following similarity measure:

$$w_{uu} : V_U \times V_U \rightarrow [0, 1] \quad (5)$$

Several implementations of w_{uu} consider the preferences expressed by two users, e.g., cosine similarity, Euclidean distance and Pearson correlation. For instance, in this last scenario we obtain the following similarity measure:

$$w_{uu}(u_j, u_k) = \frac{\sum_{l=1}^{|V_I|} (r_{j,l} - \bar{r}_j)(r_{k,l} - \bar{r}_k)}{\sqrt{\sum_{l=1}^{|V_I|} (r_{j,l} - \bar{r}_j)^2 \sum_{l=1}^{|V_I|} (r_{k,l} - \bar{r}_k)^2}} \quad (6)$$

where $\bar{r}$ is the average value of the known preferences expressed by the user. This sort of normalization is motivated by the belief that one user's scale of ratings is not

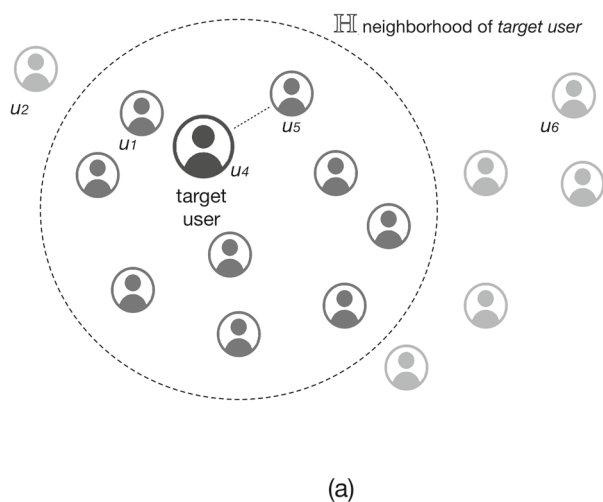
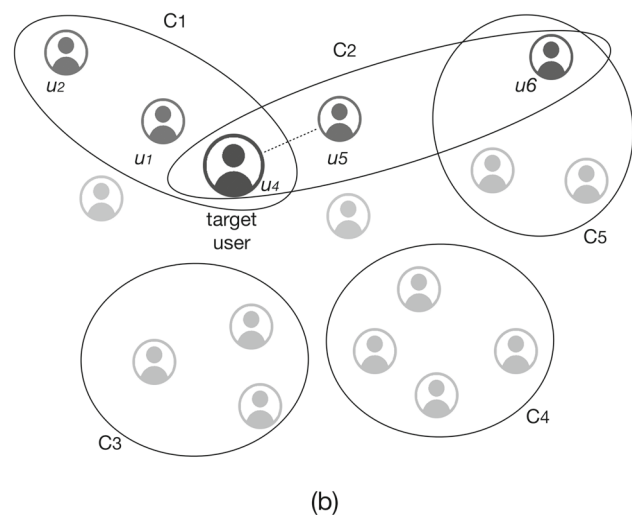


Fig. 4 A traditional Neighborhood-based CF recommendation (a), where the neighborhood of the user exploited during the recommendation is often determined by Euclidean distances. In a community-based recommendation, the neighborhood consists of a fusion of different communities of users that play a role in the recommendation process



(b). Hence, even users who are not considered really similar to the target user, and yet in the same communities, can influence the suggested items. For the sake of clarity, edges and vertices that represent items are omitted

directly comparable to another user's scale. One user might tend to rate in the high end of the scale, while another tends to use the low end.

The preference prediction of the item l for the target user u_i takes the following form:

$$\hat{r}_{i,l} = \bar{r}_i + k_0 \sum_{u_k \in \mathbb{H}_{u_i}} w_{uu}(u_i, u_k) [r_{k,l} - \bar{r}_k] \quad (7)$$

where k_0 is a normalizing factor.

Due to data sparsity of the input user-item matrix, the stage of finding similar users often fails. Two users can be considered similar only if they have rated at least few items in common. However, the items can be millions and a typical user provides preferences only about a tiny percentage of them. In addition, due to the cold start problem, the odds that another user shares a large number of rated items to a new one is small. Both data sparsity and cold start problem cause a no overlap between the preferences provided by two generic users.

Indeed, many additional social signals are not being considered with the result of often limiting the neighborhood set to few users and missing relevant resources to be recommended.

4.4.1 Social ties in the recommendation

To address the sparsity issue and improve the recommendation performances, Ma et al. [86] propose to combine the explicit social ties in the latent factor analysis by using Probabilistic MF (PMF). The assumption is that social connections affect the user's judgement of interest in items. Similarly, Liu and Lee [82] replace the neighborhood set $\mathbb{H}_u$ with the target user's friends on the social network. Of course, these attempts are effective when many explicit social ties are available, but in traditional social networks people often come together and take part to discussions and social activities because of transitory common interests or goals, e.g., football finals or movie premiers. Explicit ties between every pair of peers are not always required for backing this form of user experiences.

That being so, several approaches for the P2P and I2P link prediction have been evaluated in [97]. The authors analyzed a group of users extracted from an online vertical portal, which is considered a community. By analyzing the user activity in terms of comments, ratings and viewed resources, an implied graph of implicit ties has been derived. Measures of popularity and engagement are also considered for evaluating persons and items shared in each community. The evaluation proves how implicit social signals between community's peers and items provide a more detailed graph representation w.r.t. the one based solely on explicit ties between acquaintances or friends.

4.4.2 Social trust in the recommendation

On a different note, certain ratings may also be perceived more relevant than others depending upon who are expressing them, e.g., close friends or experts in the field under consideration. In real life, people mostly seek advice from others they know and trust. Some authors propose to make use of trust information explicitly expressed by the users and build up the neighborhood set $\mathbb{H}_u$ accordingly (e.g., see [2, 22, 50]). The rationale is that trust captures similarity between users and, therefore, a correlation measure is determinable.

There are different ways to express trust. Each user may indicate how much she considers every other trustworthy, that is, how much relevant are considered the peer's ratings. Since, this information is directly encoded in the graph-based representation of the social network, it is often named *graph-based* trust model. Assuming that the transitivity subsists, trust propagates over the social network and expands the initial $\mathbb{H}_u$ with more users that are considered trustworthy so that, even if the target user has few recorded preferences, recommendations are always computable [92]. This mechanism can also be used to infer indirect trust among participants who are not explicitly connected with a relationship. It must be said that these SRS expand the $\mathbb{H}_u$ set by disseminating trust into successive peers. But the spreading is still limited to peers close to the target user, that is, the users reachable by explicit or implicit connections. Relevant communities of peers that have the chance to provide fruitful information, but those that are located out of the scope of the local search, are just ignored.

More advanced approaches are based on MF techniques that consider both ratings and trust relationships between users [61, 87]. A social trust matrix S denotes, for each pair of users u_k and u_j , how much a user u_k trusts or knows u_j . The inferred rating matrix $\hat{R}$ is computed as follows:

$$\hat{R} = SR \quad (8)$$

that is, it only considers the trusted friends' tastes.

The principal drawback of trust-aware recommenders is that users have to state their trusted peers manually, which is perceived a burden on the normal activities. Moreover, trust relationships and explicit social ties do not necessary have the same relevance in every potential considered topic during the recommendation. If I am thinking over a movie for tonight, I would turn to my circle of cinema goer friends for suggestions ignoring more trusted ones. To overcome these issues, *interaction-based* trust model exploit the interaction data to compute trust, but ignore the characteristics of the social network structures [2].

An attempt to automatically identify and categorize trust relationships exploits once again the social signals [126].

Basically, if an explicit social tie connects two users, and if they have expressed preferences on a specific category (e.g., cars or TV shows), they belong to the same *circle*. The hypothesis is that the user may trust each friend only concerning certain item categories but not regarding others. Instead of asking for suggestions from the entire neighborhood of friends, the ratings come from the specific circle of interest.

4.5 A short review of community-based recommender systems

The principal assumption of community-based recommenders is that the affinity of the users within groups they belong to is as much relevant as the affinity of each individual in her egocentric network [56].

Once the data about users' preference ratings and social ties have been collected, three tasks have to be implemented in order to provide suggestions to the target user: (1) identifying the communities in the social network, (2) selecting communities and users that will be involved in the process, and (3) generating the recommendations.

Figure 4b shows how, after having performed a community detection on the input dataset, several clusters of users have been recognized. The C_1 and C_2 communities, which include the target user, are characterized by having a smaller number of users considered similar to one another. So instead of the whole neighborhood of users, the communities focused on specific topics and preferences can be considered for the recommendation [124].

As a result of the community detection, each user belongs to one or more communities. Depending on the specific communities and additional information chosen for the recommendation, two paradigms are feasible:

- **Single-domain recommendation:** The ranked list of suggested items are generated by analyzing a single community, which includes the target user and is assumed to be the most relevant for the recommendation.
- **Multi-domain recommendation:** The recommendation is based also on information from auxiliary or secondary communities, which in general contain the target user.

A brief survey of relevant approaches for each of the two paradigms are reported in the following sections.

4.5.1 Single-domain recommendation

Let us consider the I2P scenario. Since we are considering the users in the same community of the target user's, a very simple way is to average all the known ratings within it and choose the items with highest rank [127]. Alternatively,

a prediction based on the CF paradigm can be considered by setting $\mathbb{H}_u$ to the users belonging to the community of the target user [100, 104, 107, 111, 118]. The assumption is that preferences and interests of the community's users approximate the target user's profile. This paradigm is often named *second group* since it takes into consideration the most prominent groups of analogous users [115].

The principal advantage of this approach is to limit the number of items to consider in the analysis, and makes it easier to scale automated collaborative filtering systems to large numbers of items while maintaining reasonable prediction performance. Figure 5 consists of a rating matrix. Given the target user u_2 , the recommended items are obtained by limiting the calculus to the submatrix that represents the community C_1 , to which she belongs.

Ungar and Foster [119] and Reddy et al. [107] propose to cluster both users and items separately, so that two users who have seen or listen to similar multimedia content are grouped together, and this content is grouped based on who accesses them. The authors suppose that this approach provide more generalization beyond individual item groups, and thus should help with sparse data.

Exploiting additional sources Pham et al. [104] make use of explicit ties extracted from a snapshot of co-authorship network (namely, DBLP [120]) to cluster users and perform venue recommendation. Similarly, Fatemi and Tokarchuk [72] determine the communities by partitioning the social graph of Facebook friendships. They tackle cold-start issues by linking new users to one community by means of information that is usually readily available, and it does not depend on the domain of interest.

Similarly, in the movie domain, Fatemi and Tokarchuk [41] extract the communities from the user reviews published on Internet Movie Database (IMDb) [60] by means of the Louvain algorithm. They assume that, if many

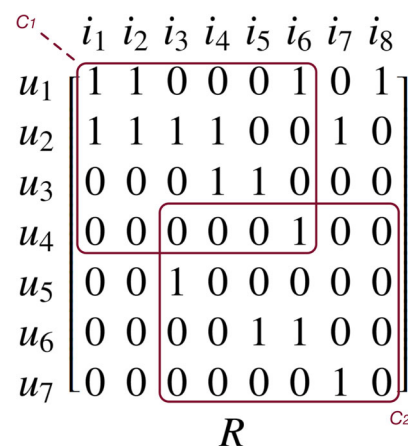


Fig. 5 Two communities of users and their ratings

users in the community C have manifested interest to i_j and i_k movies, and if i_j is in the top-ranked set, the adjacent item i_k will then be considered for recommendation. The rank of the movie in each community depends on its popularity in terms of number of reviews and on the average rating. The authors face the cold-start problem by recommending to new users the top-ranked movies from each community. When preferences of the u user are available, the *main* community is identified with the one that contains the highest number of movies previously reviewed by u , and the recommendations consist of movies adjacent to movies previously reviewed by u in the main community.

The mentioned approaches perform community detection making use of implicit and explicit social ties. Whereas explicit ties such as friendship and follow relationships can be considered stable, the graph built by implicit ties associated with user actions has a dynamic nature that makes unsupervised clustering computationally expensive if it has to be regularly updated.

Estimating the tie strength For better assessing the contribution of each community's user, additional weighting schemes can be introduced in order to explicitly represent the tie strength between users. Ma et al. [89] examine the trust relationships to highlight users considered experts in the target community. In the circumstance of cold start, the new user is being automatically connected to the identified experts by a trust relationship. The rationale is that there exists a correlation between trust and user similarity [90], so that user missing ratings of certain items are estimated by the weighted average ratings of his/her trusted neighbors who have rated the same items in the community of the user. A random walk strategy is used to explore each community and collect the trusted neighbors for users. Yin et al. [128] evaluates an activeness measure with the assumption that a person with more social ties and, therefore, higher reputation would provide more relevant contributions in a certain community.

Sentiment analysis on tweets on similar topics determines if two users have similar opinions, and Gurini et al. [52] use techniques to perform sentiment-based community detection for P2P recommendation. Opinion mining deals with computational methods in order to find, extract and systematically analyze sentiments and attitudes towards certain topics [54, 79]. While user generated content on social channels presents numerous challenges regarding systematic analysis, a number of algorithms have been proven to be applicable to SNSs [103].

One more assumption is that, if two users take actions on uncommon items, the implicit similarity between them should be considered stronger with respect to users who visit very popular items. This assumption has been evaluated in the news domain [85].

4.5.2 Multi-domain recommendation

In multi-domain recommendation users may belong to multiple communities. Sahebi and Cohen [109] introduce a $N_{|V_U|} \times N_{|C|}$ matrix where each element indicates the user membership level to a given community. The matrix provides evidence for evaluating $w_{uu}(u_j, u_k)$ according to the communities u_j and u_k belong to. The set $\mathbb{H}_u$ extends to all the users in the target user's communities, that is:

$$\mathbb{H}_u \leftarrow \bigcup_{C_j | u \in C_j} C_j \quad (9)$$

The authors focus on the cold start problem with the assumption that, if the user is new to a online service (e.g., Youtube), historical data from distinct services (e.g., Facebook) may be considered to bootstrap the recommendation algorithms. The experimental evaluation considers multiple signals, such as explicit ties (i.e., friendships), comments and ratings; each individually exploited to define communities.

Latent models Both community detection algorithms and model-based approaches, such as MF and Singular value decomposition [35], leads to significant improvements of the accuracy of the recommendations by identifying fewer numbers of relevant features and clusters of users.

In the P2P scenario, Zhao et al. [129] consider the candidates from each communities and merge them so that only the top-ranked items are kept for recommendation. They explore the problem with two latent spaces. Initially, the latent topic distribution on Twitter is modeled, and is used to group users whose probability of a given topic is above a given threshold by making use of the Latent dirichlet allocation [10]. After that, MF on each community reduces the dimensions of the user-user matrix and generates a list of candidate followees for recommendation. Li et al. [78] propose a MF model that incorporate the overlapping community information as regularization terms. They speculate that if the target user is more interested in movies than music, users belonging to the movies community should be weighed higher.

Yin et al. [128] employ probabilistic Latent Semantic Analysis (PLSA) models [55] to discover communities, so that probabilistic distributions of users and items on each community can be evaluated and used to determine the predicted ratings. If one user or item appear in many communities simultaneously, the final results are obtained by linear averaging each user-item rating determined in a community. A similar approach is taken in [130] for the P2P scenario, where users are represented by probability distributions over communities, and each community is a macro-topic of interest. The similarity measure between two users is evaluated by Kullback-Leibler divergence [26],

which quantifies in bits how close a probability distribution is to a candidate distribution. By introducing a sort of social regularization, Ma et al. [90] are able to implicitly take into account the contributions of multiple communities in the latent space of MF models, making unnecessary any form of aggregation techniques.

Das et al. [29] combine techniques such as CF, MinHash (or the min-wise independent permutations locality sensitive hashing scheme [15]) and pLSA to cluster users by analyzing the number of times a news story was covisited with the other stories in the user's click-history. In particular, MinHashing is a probabilistic clustering method that group two users with probability proportional to the overlap between the set of news that these users have visited. pLSA implements the low-dimensional representation in which users and news stories are mapped into. The authors devote a lot of effort for the implementation of a distributed algorithm for guaranteeing the scalability of the approach for the Google News online service. Shi et al. [112] use the same strategy in the movie domain, by discovering a MF-based latent space and grouping similar users with k-means clustering.

On the domain of user reviews on general consumer websites and check-ins left on social medial platforms, Cui et al. [27] examine the multinomial distributions over the topics that tags extracted from the user's reviews refer to, so that topic-based user profiles are determined. An unsupervised clustering algorithm clusters the interest topics of each user into many topic cliques. The goal of the P2P recommender is to identify the users with the highest comprehensive similarities on the obtained topic latent space, which can be implemented through the Jensen–Shannon divergence between two probability distributions, often considered in the natural language process [91].

In very large dimensional spaces with sparse vectors, the *embedding* techniques map high-dimensional vectors to low-dimensional spaces [17], making it easier to classification algorithms to compare and group together similar elements and represent non-trivial similarities between the patterns of entities in single-relational data. In [88], each user is assigned to a embedding vector where each i -dimension represents the probability that the user belongs to the i -group that clusters users with similar interests. The authors assume that, whenever the user's interest on a item is unknown, the item can still have the chance to be recommended if the user has shown interest with some other items sharing the same group with the unobserved item, or users within the same user's groups have positively rated the unobserved item.

Multiple affiliation problem and aggregation models Two communities may significantly differ in terms of cardinality,

or average popularity of their items, and the predicted ranks of the items in one community cannot be directly comparable to the ones in a different community, a problem often named *multiple affiliation problem*. Yang et al [125] characterize each community with specific characteristics, such as the strength of interaction among group members, the average number of message exchanged, and the interest conformity in terms of degree of common interest among the group. Logic regression takes into account these features to estimate the overall score and, if the same item is present in two or more communities, the Borda fuse aggregation [5, 9] promotes the broadly supported suggestions. A linear combination of recommendations coming from the target user's communities which promote items with highest predict scores is implemented in [128].

More complex models, such as *preference aggregation models* [66] and *voting schemes* [30], can be considered for aggregating several individuals' preferences into a collective preference order over the resources shared in various communities.

For instance, [123] make distinction between the effects of explicit ties and the influence that specific communities of people might have on the activity of the user. The approach is based on Graph Attention Networks (GAT) [121], a non-spectral learning method which utilizes the spatial information of the graph nodes so that neural networks can learning also input patterns represented by graph structures, such as the ones used in social network services. These convolution-style neural networks are able to assign different relevance to different nodes within a neighborhood while dealing with different sized neighborhoods, without the requirement to know the entire graph structure upfront. They are used by the authors both to characterize the user-item interaction, and incorporate information from related items, which could alleviate data sparsity problem. A multi-armed bandit model [65] determines the optimal ranking combining the outputs of neural networks.

Diversity and serendipity In order to prevent the popular *filter bubble* phenomenon, where users are trapped in a self-reinforcing cycle of opinion, never being pushed to discover alternative resources or viewpoints, *diversity* [71] and *serendipity* [70] can be explicitly modeled during the community-based recommendation. Indeed, by sifting through the resources shared in a user's communities, it is not difficult to spot items not only relevant and novel to the target user, but also significantly different from the items that the user has previously interacted with. This might improve user satisfaction with respect to the suggested items that could be interesting to users who like many items in common with the target user, which is the common assumption of CF-based approaches.

Kamahara et al. [63] explicitly address the over-specialization issue by recommending also items that are not considered similar to those already rated by the target user, but still included in the user's communities. The system builds a hierarchical model of word-word similarities that is used to group similar users according to their profiles of interest represented by a bag-of-words representation. This is a traditional content-based approach often considered in the Information Retrieval domain [45].

Expert finding As for the P2P recommendation, peculiar rating strategies have the chance to improve the accuracy of the candidate people to be suggested to the user. *Expert finding* [81] techniques can be considered for locating the users that are more skillful in a particular community.

After having clustered users in a latent space by means of rating information, Lim et al. [74] aim at promoting selected users as experts. In the music domain, an expert is considered who listens to a specific genre the most, that is, with a narrow spectrum of music diversity. Once a user asks for recommendation, the songs in her regular listening pattern are used to identify the cluster of users with similar tastes. The set of top-rated songs from the cluster's experts become candidates for the final recommendation list, which are sorted by the number of co-occurrences.

Temporal aspects Communities are continuously subjected of alterations since both user preferences, activities and items to be recommended may alter over time. It is relevant to keep track of those changes in the recommendation process, both for updating the source of information for the ranking algorithms, and for extracting relevant temporal patterns that might be of interest. As previously mentioned, the easiest way to face this aspect is to represent the social graph as a set of independent snapshots taken at different time steps. Much more difficult is to represent temporal network that explicitly include sequences of structural modifications of the network over time.

Abdrabbah et al. [1] make use of a multi-agent system that mimics the evolution of a social graph and, specifically, the creation and extinction of the communities [19]. They model the temporal changes of citizens' interests to the e-government services for recommendation. Indeed, the same service may be relevant in a certain period, circumstance and if it matches other services, but useless otherwise (e.g., a birth certificate required for school enrollment). A knowledge graph represents this information by edges that bind two services if they are frequently used by users in a certain time frame. The graph is subjected to community detection and the preferences of the target user select the closest communities from which the candidate services are collected and subjected to a traditional item-based CF.

In the specific domain of news recommendation, several temporal aspects should be carefully taken into consideration. The relevance of news article can change rapidly, going through breaking news to outdated information in few hours. In the same way, user interests on specific events might change, and may depend on different contextual factors like the time of the day, the current user activity and location [64]. For instance, [85] makes the assumption that two users who read the same news at the same time imply that they have strong similarity. A sort of forgetting phenomenon is also considered, so that user interests who do not get updated over time, will be weakened gradually, lowering their effects on the recommendation process.

In [42], temporal alterations of user interests are explicitly incorporated in community detection. A user-user graph represents implicit ties between users that show similar interests on specific items. The weight of the ties is stronger if these interests are manifested in the same timespan. A variation of the Local Fitness Maximization (LFM) algorithm [73] is used for community detection. The recommendations are based on the outcomes of association rule mining [3], which determines the most likely items that are considered by the users in a certain community after having considered one or more items in the past. The same mining approach has been followed by [107].

4.5.3 Final remarks

As the comparison of the proposed community-based recommenders reflects (Table 3), the combinations of techniques implemented in each system is pretty much a mixed picture. This is a clear sign that additional investigation is required to reach a broader consensus on the most effective and accurate systems. Nevertheless, the following observations can be inferred:

- While several approaches have been successfully experimented in the movie domain, given the availability of popular public large datasets, standard benchmarks on multiple domains that allow us to make a direct comparison of the performances are yet to be defined. Indeed, one approach might prove its effectiveness in one specific domain, whereas on other features and patterns that characterize different domains might fail to reach similar performance.
- Several approaches leverage prior knowledge to define latent models that represent both users and items. This step increases the expressive power of the internal representations and, at the same time, reduce the complexity of real-world data, often complex and high-dimensional.

Table 3 Comparison between surveyed community-based recommender systems

References	Link prediction	Domain of interest	Input social signals	Community detection	Ranking Techniques
Single-domain recommendation:					
Ungar and Foster [119]	I2P	Movies	R	UC	LC
O'Connor and Herlocker [100]	I2P	Music	R	UC	CF
Reddy et al. [107]	I2P	Movies	R	DBG	AR
Sarwar et al. [111]	I2P	Movies	R	LCD	CF
Truong et al. [118]	I2P	Movies	R	UC	CF
Pham et al. [104]	I2P	POI	ET	LCD,UC	CF
Yanxiang et al. [127]	I2P	Movies	R	UC	LC
Fatemi and Tokarchuk [41]	I2P	Movies	R,IT	LCD	LC
Ma et al. [89]	I2P	e-com	R,T	SVD	CF
Lalwani et al. [72]	I2P	Movies	R	LCD	CF
Multi-domain recommendation:					
Kamahara et al. [63]	I2P	TV programs	IT	UC	CB
Das et al. [29]	I2P	News	IT	pLSA	CF
Zhou et al. [130]	P2P	Links	IT	LCD	KL
Sahebi and Cohen [109]	I2P	Books,Movies	R,IT	LCD	CF
Xu et al. [124]	I2P	Movies,Music	R	UC	CF,LC
Zhao et al. [129]	P2P	Twitter	ET	LDA	MF,LC
Gurini et al. [52]	P2P	Tweets	IT	UC,SA	LC
Lee and Lee [74]	I2P	Music	IT	SVD,UC	LC
Ma et al. [90]	I2P	Various	R,ET	UC	MF
Meilian et al. [85]	I2P	News	IT	UC	LC
Yin et al. [128]	I2P	e-com	R,IT	pLSA	MF,LC
Feng et al. [42]	I2P	Movies	R	LFM,AR	LC
Li et al. [78]	I2P	POI,Movies	R,ET	MF	LC
Shi et al. [112]	I2P	Movies	R	MF,UC	LC
Abdrabbah et al. [1]	I2P	e-gov	R	ACD	CF
Yang et al. [125]	I2P	Movies	IT	EG	TS
Ma et al. [88]	I2P	POI,Movies	R	E	NN
Wu et al. [123]	I2P	e-com,News	IT	NN	MAB
Cui et al. [27]	P2P	Reviews	IT	UC	JS

(ACD) Community detection based on Multi-agent system, (AR) Association rule-based recommendation, (C) Commenting activity, (CB) Content-based ranking, (DBG) Community detection based on dense bipartite graph, (E) Embeddings, (EG) Communities corresponds to social network's groups, (ET) Explicit social ties, (IT) Implicit social ties, (JS) User-user similarity based on Jensen–Shannon divergence, (KL) User-user similarity based on Kullback–Leibler divergence, (LC) Linear combination, (LCD) Modularity-based community detection, (LDA) Latent Dirichlet Allocation, (LFM) Local Fitness Maximization, (MAB) Multi-armed bandit model, (MF) Matrix factorization, (ML) Link prediction based on Maximum likelihood, (PLSA) pLSA-based model, (PM) Link prediction based on Probabilistic models, (POI) Points of interest recommendation, (R) Rating/liking activity, (SA) Sentiment analysis, (SB) Similarity-based link prediction, (SVD) Singular value decomposition, (T) Trust relationships, (TS) Ranking based on user-item tie strength, (UC) Common unsupervised clustering algorithm

- In recent years a few attempts based on Deep learning architecture have proved their potential in community-based recommendation.
- In almost all the considered domains, transactional or feedback data might be sparse and insufficient for identifying neighbors, and this is a major issue limiting the quality of recommendations and the applicability of traditional collaborative filtering in general.
- The literature prove how implicitly derived clusters of similar users or items, or communities explicitly defined by users on social network services support the improvements of the recommendation accuracy.
- As other side of the same coin, by clustering similar users and items it is also possible to explicitly face scalability issues and speed up the computations. But very few researchers have evaluated the proposed

systems on large real-data scenarios, or proposed to rely on distributed computing mechanisms and to execute the calculations on multiple servers in parallel.

- More recently, we observe papers on important beyond-accuracy quality aspects like diversity, novelty and serendipity. It reflects a similar trend followed in the broader recommender system community.
- It is empirically proven that additional sources of data provide a way to increment the performance, especially when the user-item matrix of ratings or preferences lack of adequate coverage. But it is not clear if these sources should be leveraged during community detection or combined in the ranking phase.

5 Conclusions and future directions

It has been reported how some of the current recommendation systems already benefit from detecting real or virtual communities of users extracted from social platforms.

Multiple steps are required to combine community detection techniques in the recommendation process, which make this goal theoretically and practically challenging. Several critical scenarios and research challenges have to be addressed to see community-based recommender systems make headway, such as:

- Recommender systems benefit by including additional information from auxiliary sources, or even from within the social network. For instance, an e-commerce service may observe ratings from the book store to improve the recommendation accuracy in the movie store by supposing that a correlation exists between the two domains. This form of *cross-domain recommendation* can be extended by comparing behaviors and feedbacks of users in two or more communities in order to infer patterns helpful for the recommendation [18].
- Potential drifts of interest, alterations of the relationships between peers and the constantly change of the popularity of shared items are all factors that affect the recommendation accuracy. For instance, Nepal et al. [96, 97] and Yin et al. [128] assume that recent interactions are more relevant than ones occurred a long time ago, and they weight the respective contribution accordingly. *Evolving social graph clustering* or *dynamic community detection* techniques [20] that take into account these temporal dimensions have been substantially ignored to date.
- The survey shows how a prevalent model that fits all the categories of recommended items, and both the I2P vs P2P scenarios, is still to be determined. If the benefits of community detection in the recommendation process has been empirically proven several times on

real-world datasets, guiding principles to understand which approach is more suitable in a specific scenario are yet to be developed.

- The study reveals difficulties in obtaining realistic and accurate lab-based evaluations for assessing the recommendation accuracy due to the scarcity of large datasets that include explicit and implicit social signals. Hence, a comparative evaluation for estimating the accuracy of different community-based RS on the same input dataset is not available.
- Recent advancements in deep learning technologies provide new tools for assign compact representation of complex graph structured data in the form of embeddings (e.g., *graph2vec* [94]). In addition, recurrent neural networks are now popular techniques for session-based recommendation and, more in general, for taking into account temporal aspects of user preferences [33]. It is logical that future community-based recommender systems may exploit these tools for improving the understanding of the collected data and improve the accuracy.
- Session-based recommender systems is an emerging research topic in the recommendation domain [62], but none of the surveyed community-based recommender make explicitly reference to short-term transactional patterns or sessions, identifying significant patterns and the user preference shift from one transaction to another.
- Recommenders exploit the collective behavior of users in order to deliver highly targeted recommendations. Empirical evaluations are mostly based on samples of the whole population, without considering statistical biases, including the ones generated by the sampling process, validity, completeness, noise, or spam. There are cases where recommenders may amplify such biases, leading to the phenomenon of bias disparity. While lately several attempts have been proposed to reduce this phenomenon [37], community-based recommender systems designed with societal values such as fairness, accountability, and transparency in mind are yet to be investigated.

References

1. Abdrabbah SB, Ayachi R, Amor NB (2016) A dynamic community-based personalization for e-government services. In: Proceedings of 9th international conference on theory and practice of electronic governance, ICEGOV '15–16. ACM, New York, pp 258–265
2. Aggarwal CC (2016) Social and trust-centric recommender systems. Springer International Publishing, Cham, pp 345–384
3. Agrawal R, Imieliński T, Swami A (1993) Mining association rules between sets of items in large databases. SIGMOD Rec 22(2):207–216

4. Arazy O, Kumar N, Shapira B (2009) Improving social recommender systems. *IT Prof* 11(4):38–44
5. Aslam JA, Montague M (2001) Models for metasearch. In: *Proceedings of 24th annual international ACM SIGIR conference on research and development in information retrieval, SIGIR '01*. ACM, New York, pp 276–284
6. Azaouzi M, Rhouma D, Romdhane LB (2019) Community detection in large-scale social networks: state-of-the-art and future directions. *Social Netw Anal Min* 9(1):23
7. Backstrom L, Huttenlocher D, Kleinberg J, Lan X (2006) Group formation in large social networks: membership, growth, and evolution. In: *Proceedings of 12th ACM SIGKDD international conference on knowledge discovery and data mining, KDD '06*. ACM, New York, pp 44–54
8. Bai X, Wang M, Lee I, Yang Z, Kong X, Xia F (2019) Scientific paper recommendation: a survey. *IEEE Access* 7:9324–9339
9. Baltrunas L, Makcinskas T, Ricci F (2010) Group recommendations with rank aggregation and collaborative filtering. In: *Proceedings of the fourth ACM conference on recommender systems, RecSys '10*. Association for Computing Machinery, New York, pp 119–126
10. Blei DM, Ng AY, Jordan MI (2003) Latent dirichlet allocation. *J Mach Learn Res* 3:993–1022
11. Blondel VD, Guillaume J-L, Lambiotte R, Lefebvre E (2008) Fast unfolding of communities in large networks. *J Stat Mech* 2008:1–12
12. Boehmer J, Jung Y, Wash R (2015) e-commerce recommender systems. *The International Encyclopedia of Digital Communication and Society*, pp 1–8
13. Boettcher S, Percus AG (2001) Optimization with extremal dynamics. *Phys Rev Lett* 86:5211–5214
14. Boratto L (2016) Group recommender systems: state of the art, emerging aspects and techniques, and research challenges. In: Ferro N, Crestani F, Moens M-F, Mothe J, Silvestri F, Di Nunzio GM, Hauff C, Silvello G (eds) *Advances in information retrieval: 38th European conference on IR research, ECIR 2016, Padua, Italy, March 20–23, 2016. Proceedings*. Springer International Publishing, pp 889–892
15. Broder AZ (1997) On the resemblance and containment of documents. In: *Proceedings. Compression and complexity of sequences 1997*, pp 21–29
16. Bu Z, Wu Z, Cao J, Jiang Y (2016) Local community mining on distributed and dynamic networks from a multiagent perspective. *IEEE Trans Cybern* 46(4):986–999
17. Camacho-Collados J, Pilehvar MT (2018) From word to sense embeddings: a survey on vector representations of meaning. *J Artif Int Res* 63(1):743–788
18. Cantador I, Fernández-Tobías I, Berkovsky S, Cremonesi P (2015) Cross-domain recommender systems. In: Ricci F, Rokach L, Shapira B (eds) *Recommender systems handbook*. Springer, pp 919–959
19. Cazabet R, Amblard F (2011) Simulate to detect: a multi-agent system for community detection. In: *2011 IEEE/WIC/ACM international conferences on web intelligence and intelligent agent technology, vol 2*, pp 402–408
20. Cazabet R, Amblard F (2014) Dynamic community detection. In: Alhajj R, Rokne J (eds) *Encyclopedia of social network analysis and mining*. Springer, New York, pp 404–414
21. Chakrabarti D, Kumar R, Tomkins A (2006) Evolutionary clustering. In: *Proceedings of 12th ACM SIGKDD international conference on knowledge discovery and data mining, KDD '06*. ACM, New York, pp 554–560
22. Chen J, Geyer W, Dugan C, Muller M, Guy I (2009) Make new friends, but keep the old: recommending people on social networking sites. In: *Proceedings of 27th international conference on human factors in computing systems, CHI '09*. ACM, New York, pp 201–210
23. Chi Y, Song X, Zhou D, Hino K, Tseng BL (2007) Evolutionary spectral clustering by incorporating temporal smoothness. In: *Proceedings of 13th ACM SIGKDD international conference on knowledge discovery and data mining, KDD '07*. ACM, New York, pp 153–162
24. Chiericetti F, Dasgupta A, Kumar R, Lattanzi S, Sarlós T (2016) On sampling nodes in a network. In: *Proceedings of 25th international conference on World Wide Web, WWW '16*. Republic and Canton of Geneva, Switzerland, pp 471–481. *International World Wide Web Conferences Steering Committee*
25. Clauset A, Newman MEJ, Moore C (2004) Finding community structure in very large networks. *Phys Rev E* 70:1–6
26. Cover TM, Thomas JA (2006) *Elements of information theory*, 2nd edn. Wiley, New York
27. Cui L, Wu J, Pi D, Zhang P, Kennedy P (2020) Dual implicit mining-based latent friend recommendation. *IEEE Trans Syst Man Cybern Syst* 50(5):1663–1678
28. Dara S, Chowdary CR, Kumar C (2020) A survey on group recommender systems. *J Intell Inf Syst* 54(2):271–295
29. Das AS, Datar M, Garg A, Rajaram S (2007) Google news personalization: scalable online collaborative filtering. In: *Proceedings of the 16th international conference on World Wide Web, WWW, '07*. Association for Computing Machinery, New York, pp 271–280
30. Das J, Mukherjee P, Majumder S, Gupta P (2014) Clustering-based recommender system using principles of voting theory. In: *2014 International conference on contemporary computing and informatics (IC3I)*, pp 230–235
31. Datasift (2020) <http://datasift.com>. Accessed: 30 May 2020
32. Deerwester S, Dumais ST, Furnas GW, Landauer TK, Harshman R (1990) Indexing by latent semantic analysis. *J Am Soc Inf Sci* 41(6):391–407
33. Devooght R, Bersini H (2017) Long and short-term recommendations with recurrent neural networks. In: *Proceedings of the 25th conference on user modeling, adaptation and personalization, UMAP '17*. Association for Computing Machinery, New York, pp 13–21
34. Dou Y, Yang H, Deng X (2016) A survey of collaborative filtering algorithms for social recommender systems. In: *2016 12th International conference on semantics, knowledge and grids (SKG)*, pp 40–46
35. Douglas EP (2009) *Clustering datasets with singular value decomposition*. PhD thesis, College of Charleston
36. Duch J, Arenas A (2005) Community detection in complex networks using extremal optimization. *Phys Rev E* 72:027104
37. Edizel B, Bonchi F, Hajian S, Panisson A, Tassa T (2020) Fairesys: mitigating algorithmic bias in recommender systems. *Int J Data Sci Anal* 9(2):197–213
38. Fan W, Yeung A (2014) Incorporating profile information in community detection for online social networks. *Phys A: Stat Mech Appl* 405:226–234
39. Fani H, Bagheri E, Zarrinkalam F, Zhao X, Du W (2018) Finding diachronic like-minded users. *Comput Intell* 34(1):124–144
40. Fani H, Jiang E, Bagheri E, Al-Obeidat F, Du W, Kargar M (2020) User community detection via embedding of social network structure and temporal content. *Inf Process Manag* 57(2):102056
41. Fatemi M, Nadia L (2013) A community based social recommender system for individuals & groups. In: *2013 International conference on social computing (SocialCom)*, pp 351–356
42. Feng H, Tian J, Wang HJ, Li M (2015) Personalized recommendations based on time-weighted overlapping community

- detection. *Inf Manag* 52(7):789–800. Novel applications of social media analytics
43. Ferrara E (2012) Community structure discovery in Facebook. *IJSNM* 1(1):67–90
44. Fortunato S (2010) Community detection in graphs. *Phys Rep* 486(3–5):75–174
45. Gauch S, Speretta M, Chandramouli A, Micarelli A (2007) User profiles for personalized information access. Springer, Berlin, pp 54–89
46. Gauch S, Speretta M, Chandramouli A, Micarelli A (2007) User profiles for personalized information access. In: Brusilovsky P, Kobsa A, Nejdl W (eds) *The adaptive web*. Springer, Berlin, pp 54–89
47. Girvan M, Newman MEJ (2002) Community structure in social and biological networks. *Proc Natl Acad Sci* 99(12):7821–7826
48. Goldberg D, Nichols D, Oki BM, Terry D (1992) Using collaborative filtering to weave an information tapestry. *Commun ACM* 35(12):61–70
49. Granovetter MS (1973) The strength of weak ties. *Am J Sociol* 78(6):1360–1380
50. Groh G, Birnkammerer S, Köllhofer V (2012) Social recommender systems. In: *Recommender systems for the social web*, vol 32. Springer, Berlin, pp 3–42
51. Guimera R, Amaral LAN (2005) Functional cartography of complex metabolic networks. *Nature* 433(7028):895–900
52. Gurini DF, Gasparetti F, Micarelli A, Sansonetti G (2014) iscur: interest and sentiment-based community detection for user recommendation on Twitter. In: Dimitrova V, Kuflik T, Chin D, Ricci F, Dolog P, Houben G-J (eds) *UMAP*, volume 8538 of *Lecture Notes in Computer Science*. Springer, pp 314–319
53. Harper MF, Konstan JA (2015) The movielens datasets: history and context. *ACM Trans Interact Intell Syst* 5(4):1–19
54. Hemmatian F, Sohrabi MK (2019) A survey on classification techniques for opinion mining and sentiment analysis. *Artif Intell Rev* 52(3):1495–1545
55. Hofmann T (1999) Probabilistic latent semantic indexing. In: *Proceedings of the 22nd annual international ACM SIGIR conference on research and development in information retrieval, SIGIR '99*. Association for Computing Machinery, New York, pp 50–57
56. Hogg T (2010) Inferring preference correlations from social networks. *Electron Commerce Res Appl* 9(1):29–37. Special Issue: Social Networks and Web 2.0
57. Hu Y, Koren Y, Volinsky C (2008) Collaborative filtering for implicit feedback datasets. In: *Proceedings of 2008 eighth IEEE international conference on data mining, ICDM '08*. IEEE Computer Society, Washington, DC, pp 263–272
58. Hu Z, Yao J, Cui B, Xing E (2015) Community level diffusion extraction. In: *Proceedings of the 2015 ACM SIGMOD international conference on management of data, SIGMOD '15*. Association for Computing Machinery, New York, pp 1555–1569
59. Twitter Inc (2020) Twitter for developers. <https://developer.twitter.com/>. Accessed: 30 May 2020
60. Internet movie database. <https://www.imdb.com>, 2020. Accessed: 30 May 2020
61. Jamali M, Ester M (2010) A matrix factorization technique with trust propagation for recommendation in social networks. In: *Proceedings of fourth ACM conference on recommender systems, RecSys '10*. ACM, New York, pp 135–142
62. Jannach D, Ludewig M, Lerche L (2017) Session-based item recommendation in e-commerce: on short-term intents, reminders, trends and discounts. *User Model User-Adapted Interact* 27(3):351–392
63. Kamahara J, Asakawa T, Shimojo S, Miyahara H (2005) A community-based recommendation system to reveal unexpected interests. In: *11th International multimedia modelling conference*, pp 433–438
64. Karimi M, Jannach D, Jugovac M (2018) News recommender systems—survey and roads ahead. *Inf Process Manag* 54(6):1203–1227
65. Katehakis MN, Veinott AF (1987) The multi-armed bandit problem: decomposition and computation. *Math Oper Res* 12:262–268
66. Sen A, Arrow KJ, Suzumura K (eds) (2011) *Handbook of social choice and welfare*. Handbook of Social Choice and Welfare. Elsevier
67. Khan MM, Ibrahim R, Ghani I, systems (2017) Cross domain recommender: a systematic literature review. 50(3). *ACM Comput Surv* 50(3):1–34
68. Kleinberg J (2000) The small-world phenomenon: an algorithmic perspective. In: *Proceedings of the thirty-second annual ACM symposium on theory of computing, STOC '00*. Association for Computing Machinery, New York, pp 163–170
69. Koren Y, Bell R, Volinsky C (2009) Matrix factorization techniques for recommender systems. *Computer* 42(8):30–37
70. Kotkov D, Wang S, Veijalainen J (2016) A survey of serendipity in recommender systems. *Knowl-Based Syst* 111:180–192
71. Kunaver M, Požrl T (2017) Diversity in recommender systems—a survey. *Knowl-Based Syst* 123:154–162
72. Lalwani D, Somayajulu DVLN, Radha Krishna P (2015) A community driven social recommendation system. In: *2015 IEEE International conference on big data (big data)*, pp 821–826
73. Lancichinetti A, Fortunato S, Kertész J (2009) Detecting the overlapping and hierarchical community structure in complex networks. *New J Phys* 11(3):033015
74. Lee K, Lee K (2014) Using dynamically promoted experts for music recommendation. *IEEE Trans Multimed* 16(5):1201–1210
75. Leskovec J, Faloutsos C (2006) Sampling from large graphs. In: *Proceedings of 12th ACM SIGKDD international conference on knowledge discovery and data mining, KDD '06*. ACM, New York, pp 631–636
76. Levin DZ, Cross R (2004) The strength of weak ties you can trust: the mediating role of trust in effective knowledge transfer. *Manag Sci* 50(11):1477–1490
77. Li L, Tang XJ (2016) A solution to the cold-start problem in recommender systems based on social choice theory. In: Lavangnananda K, Phon-Amnuaisuk S, Engchuan W, Chan JH (eds) *The 19th Asia Pacific symposium, IES 2015, Bangkok, Thailand, 2015*. Proceedings. Springer International Publishing, pp 267–279
78. Li H, Wu D, Tang W, Mamoulis N (2015) Overlapping community regularization for rating prediction in social recommender systems. In: *Proceedings of 9th ACM conference on recommender systems, RecSys '15*. ACM, New York, pp 27–34
79. Li Z, Fan Y, Jiang B, Lei T, Liu W (2019) A survey on sentiment analysis and opinion mining for social multimedia. *Multimed Tools Appl* 78(6):6939–6967
80. Liben-Nowell D, Kleinberg J (2003) The link prediction problem for social networks. In: *Proceedings of twelfth international conference on information and knowledge management, CIKM '03*. ACM, New York, pp 556–559
81. Lin S, Hong W, Wang D, Li T (2017) A survey on expert finding techniques. *J Intell Inf Syst* 49(2):255–279
82. Liu F, Lee HJ (2010) Use of social network information to enhance collaborative filtering performance. *Expert Syst Appl* 37(7):4772–4778
83. Liu J, Aggarwal C, Han J (2015) On integrating network and community discovery. In: *Proceedings of eighth ACM*

- international conference on web search and data mining, WSDM '15. ACM, New York, pp 117–126
84. Liu H, Yang F, Liu D (2016) Genetic algorithm optimizing modularity for community detection in complex networks. In: 2016 35th Chinese control conference (CCC), pp 1252–1256
85. Lu M, Qin Z, Cao Y, Liu Z, Wang M (2014) Scalable news recommendation using multi-dimensional similarity and jaccard-kmeans clustering. *J Syst Softw* 95:242–251
86. Ma H, Yang H, Lyu MR, Sorec IK (2008) Social recommendation using probabilistic matrix factorization. In: Proceedings of 17th ACM conference on information and knowledge management, CIKM '08. ACM, New York, pp 931–940
87. Ma H, King I, Lyu MR (2009) Learning to recommend with social trust ensemble. In: Proceedings of 32nd international ACM SIGIR conference on research and development in information retrieval, SIGIR '09. ACM, New York, pp 203–210
88. Ma J, Wen J, Zhong M, Liu L, Li C, Chen W, Yang Y, Tu H, Li X (2019) Dbrec: dual-bridging recommendation via discovering latent groups. In: Proceedings of the 28th ACM international conference on information and knowledge management, CIKM '19. Association for Computing Machinery, New York, pp 1513–1522
89. Ma X, Lu H, Gan Z (2014) Improving recommendation accuracy by combining trust communities and collaborative filtering. In: Proceedings of 23rd ACM international conference on conference on information and knowledge management, CIKM '14. ACM, New York, pp 1951–1954
90. Ma X, Lu H, Gan Z, Ma Y (2014) Improving recommendation accuracy with clustering-based social regularization. In: Chen L, Jia Y, Sellis T, Liu G (eds) Web technologies and applications. Springer International Publishing, Cham, pp 177–188
91. Manning CD, Schütze H (1999) Foundations of statistical natural language processing. MIT Press, Cambridge
92. Massa P, Avesani P (2007) Trust-aware recommender systems. In: Proceedings of 2007 ACM conference on recommender systems, RecSys '07. ACM, New York, pp 17–24
93. Mu R (2018) A survey of recommender systems based on deep learning. *IEEE Access* 6:69009–69022
94. Narayanan A, Chandramohan M, Venkatesan R, Chen L, Liu Y, Jaiswal S (2017) graph2vec: learning distributed representations of graphs. CoRR, arXiv:1707.05005
95. Nazi A, Zhou Z, Thirumuruganathan S, Zhang N, Das G (2015) Walk, not wait: faster sampling over online social networks. *Proc VLDB Endow* 8(6):678–689
96. Nepal S, Paris C, Bista SK (2012) Srec: a social behaviour based recommender for online communities. In: Herder E, Yacef K, Chen L, Weibelzahl S (eds) Workshop and poster proceedings of 20th conference on user modeling, adaptation, and personalization, Montreal, Canada, July 16–20, 2012, volume 872 of CEUR Workshop Proceedings. CEUR-WS.org
97. Nepal S, Paris C, Pour PA, Freyne J, Bista SK (2015) Interaction-based recommendations for online communities. *ACM Trans Internet Technol* 15(2):6:1–6:21
98. Newman MEJ (2003) Fast algorithm for detecting community structure in networks. *Phys Rev E* 69:1–5
99. Newman M, Girvan M (2004) Finding and evaluating community structure in networks. *Phys Rev E* 69(2):026113
100. O'Connor M, Herlocker JL (1999) Clustering items for collaborative filtering. In: Proceedings of ACM SIGIR workshop on recommender systems algorithms and evaluation, Berkeley
101. Pan R, Zhou Y, Cao B, Liu NN, Lukose R, Scholz M, Yang Q (2008) One-class collaborative filtering. In: Proceedings of 2008 Eighth IEEE international conference on data mining, ICDM '08. IEEE Computer Society, Washington, DC, pp 502–511
102. Pennock DM, Horvitz E, Giles CL (2000) Social choice theory and recommender systems: analysis of the axiomatic foundations of collaborative filtering. In: Proceedings of seventeenth national conference on artificial intelligence and twelfth conference on innovative applications of artificial intelligence. AAAI Press, pp 729–734
103. Petz G, Karpowicz M, Fürschuß H, Auinger A, Stříteský V, Holzinger A (2014) Computational approaches for mining user's opinions on the web 2.0. *Inf Process Manag* 50(6):899–908
104. Pham MC, Cao Y, Klamra R, Jarke M (2011) A clustering approach for collaborative filtering recommendation using social network analysis. *J Univ Comput Sci* 17(4):583–604
105. Pizzato L, Rej T, Akehurst J, Koprinska I, Yacef K, Kay J (2013) Recommending people to people: the nature of reciprocal recommenders with a case study in online dating. *User Model User-Adapted Interact* 23(5):447–488
106. Raskutti B, Kowalczyk A (2004) Extreme re-balancing for svms A case study. *SIGKDD Explor Newsl* 6(1):60–69
107. Reddy PK, Kitsuregawa M, Sreekanth P, Rao SS (2002) A graph based approach to extract a neighborhood customer community for collaborative filtering. In: Proceedings of second international workshop on databases in networked information systems, DNIS '02. Springer, London, pp 188–200
108. Ribeiro B, Towsley D (2010) Estimating and sampling graphs with multidimensional random walks. In: Proceedings of 10th ACM SIGCOMM conference on internet measurement, IMC '10. ACM, New York, pp 390–403
109. Sahebi S, Cohen W (2011) Community-based recommendations: a solution to the cold start problem. In: Workshop on recommender systems and the social web (RSWEB), held in conjunction with ACM RecSys 2011
110. Sahoo AK, Pradhan C, Barik RK, Dubey H (2019) Deepreco: deep learning based health recommender system using collaborative filtering. *Computation* 7(2):25
111. Sarwar BM, Karypis G, Konstan J, Riedl J (2002) Recommender systems for large-scale E-commerce: scalable neighborhood formation using clustering. In: 5th International conference on computer information technology (ICCIT)
112. Shi J, Bin Wu, Lin X (2015) A latent group model for group recommendation. In: 2015 IEEE International conference on mobile services, pp 233–238
113. Shokeen J, Rana C (2020) A study on features of social recommender systems. *Artif Intell Rev* 53(2):965–988
114. SocialGist Socialgist. <https://socialgist.com>. Accessed: 30 May 2020. 2020
115. Son LH (2016) Dealing with the new user cold-start problem in recommender systems: a comparative review. *Inf Syst* 58:87–104
116. Statista Number of social media users worldwide from 2010 to 2023 (in billions), December 2019. Last retrieved: 2 April 2020
117. Tchuente D, Canut M-F, Baptiste-Jessel N, Peninou A, Sedes F (2012) A community based algorithm for deriving users' profiles from egocentrics networks. In: Proceedings of 2012 international conference on advances in social networks analysis and mining (ASONAM 2012), ASONAM '12. IEEE Computer Society, Washington, DC, pp 266–273
118. Truong KQ, Ishikawa F, Honiden S (2007) Improving accuracy of recommender system by item clustering. *IEICE - Trans Inf Syst* E90-D(9):1363–1373
119. Ungar LH, Foster DP (1998) Clustering methods for collaborative filtering. In: Workshop on recommender systems at the 15th national conference on artificial intelligence (AAAI'98). AAAI Press, Madison, pp 112–125
120. Germany University of Trier (2020) Database systems and logic programming. <https://dblp.org/>. Accessed: 30 May 2020

121. Veličković P, Cucurull G, Casanova A, Romero A, Liò P, Bengio Y (2018) Graph attention networks. In: International conference on learning representations
122. Wasserman S, Faust K (1994) Social network analysis: methods and applications. In: Granovetter M (ed) Social network analysis: methods and applications (structural analysis in the social sciences). Cambridge University Press
123. Wu Q, Zhang H, Gao X, He P, Weng P, Gao H, Chen G (2019) Dual graph attention networks for deep latent representation of multifaceted social effects in recommender systems. In: The World Wide Web conference, WWW '19. Association for Computing Machinery, New York, pp 2091–2102
124. Xu B, Bu J, Chen C, Cai D (2012) An exploration of improving collaborative recommender systems via user-item subgroups. In: Proceedings of 21st international conference on World Wide Web, WWW '12. ACM, New York, pp 21–30
125. Yang C, Zhou Y, Chen L, Zhang X, Chiu DM (2016) Social-group-based ranking algorithms for cold-start video recommendation. *Int J Data Sci Anal* 1(3):165–175
126. Yang X, Steck H, Liu Y (2012) Circle-based recommendation in online social networks. In: Proceedings of 18th ACM SIGKDD international conference on knowledge discovery and data mining, KDD '12. ACM, New York, pp 1267–1275
127. Yanxiang L, Deke G, Fei C, Honghui C (2013) User-based clustering with top-n recommendation on cold-start problem. In: 2013 Third international conference on intelligent system design and engineering applications, pp 1585–1589
128. Yin B, Yang Y, Liu W (2014) Exploring social activeness and dynamic interest in community-based recommender system. In: Proceedings of 23rd international conference on World Wide Web, WWW '14 Companion. ACM, New York, pp 771–776
129. Zhao G, Lee ML, Hsu W, Chen W, Hu H (2013) Community-based user recommendation in uni-directional social networks. In: Proceedings of 22nd ACM international conference on information & knowledge management, CIKM '13. ACM, New York, pp 189–198
130. Zhou TC, Ma H, Lyu MR, King I (2010) Userrec: a user recommendation framework in social tagging systems. In: Proceedings of the twenty-fourth AAAI conference on artificial intelligence, AAAI'10. AAAI Press, pp 1486–1491
131. Zigran S, Bronstein J (2019) Help is where you find it: the role of weak ties networks as sources of information and support in virtual health communities. *J Assoc Inf Sci Technol* 70(2):130–139

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